Quality control of sequencing data

Introduction to Bioinformatics · NGS day 1

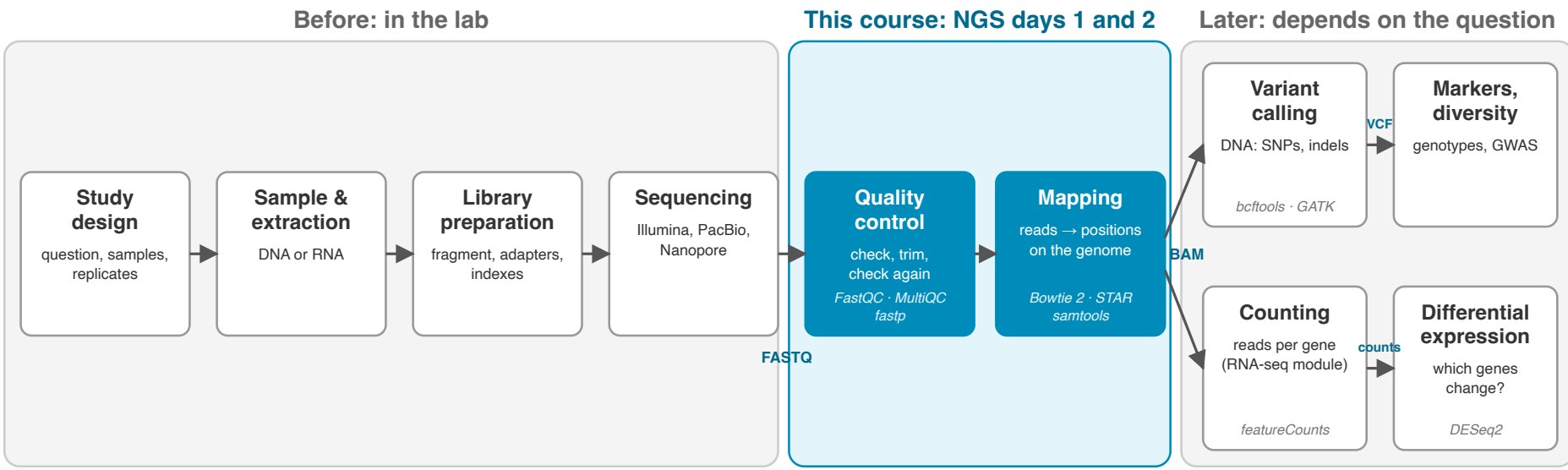
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This lecture

After this lecture you can:

1. Explain why QC of sequencing data is necessary
2. Interpret FastQC reports and their warnings
3. Decide what action, if any, to take based on QC results

Where we are in the workflow



Why quality control?

From plant to reads

> Plant tissue → DNA extraction → Library preparation → Sequencing → FASTQ files

Each step can introduce technical effects that later show up in QC reports.

We sequence to answer a **biological question**: which variants, which alleles, which genes are expressed.

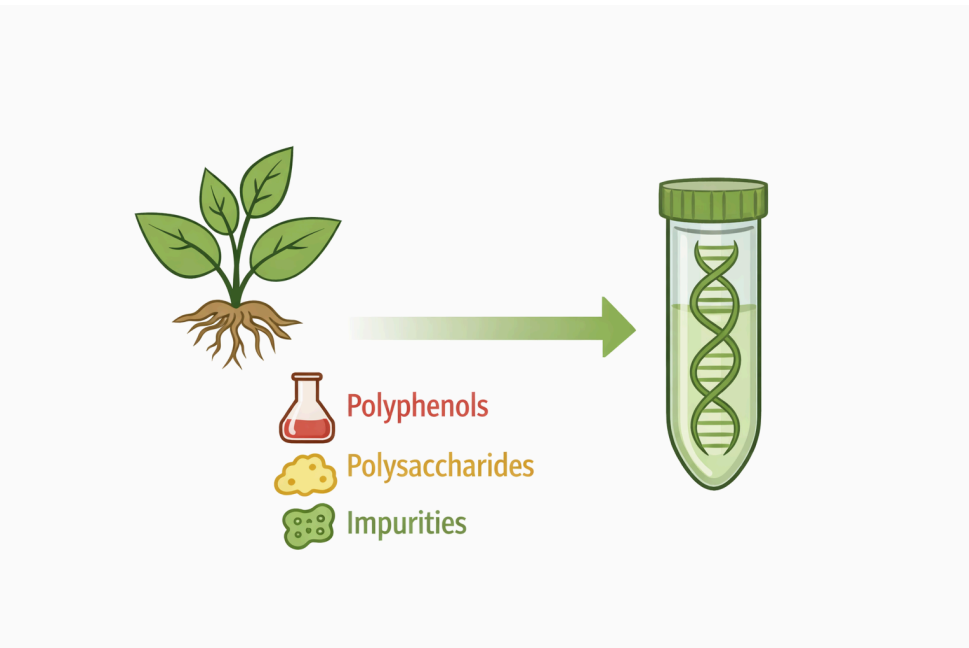
All of these assume that **the sequencing data reflects the real biology of the plant.** QC checks whether that assumption is reasonable.

What can go wrong?

Technical issue		Biological consequence
Low base quality	→	False SNPs
Adapter contamination	→	Artificial variants
GC bias	→	Skewed allele frequencies
High duplication	→	Overconfidence in genotypes

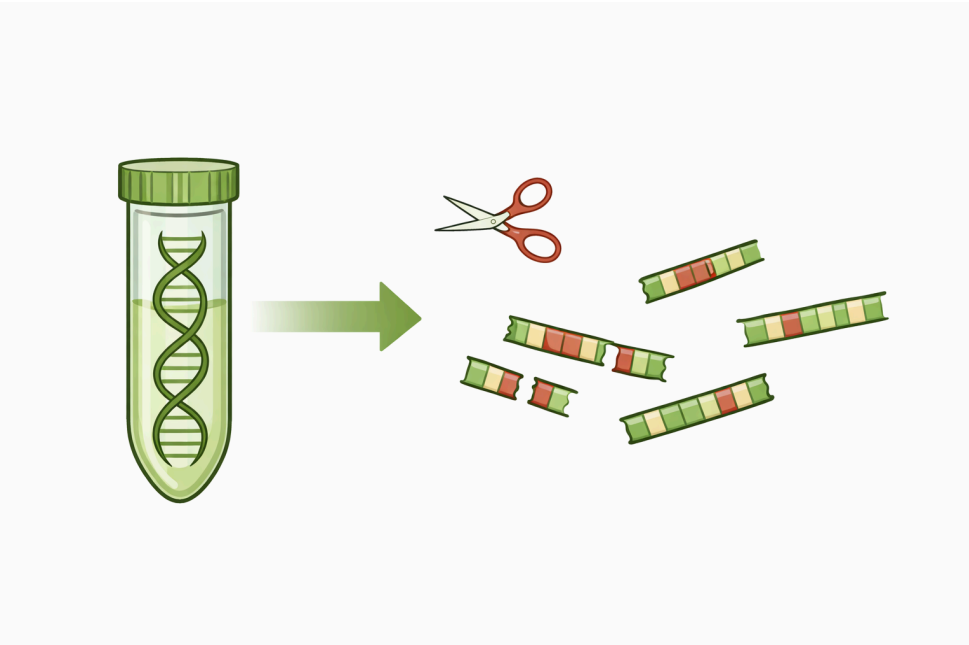
In a breeding context this could mean choosing the wrong markers, missing true marker-trait associations, misjudging genetic diversity.

Step 1: DNA extraction



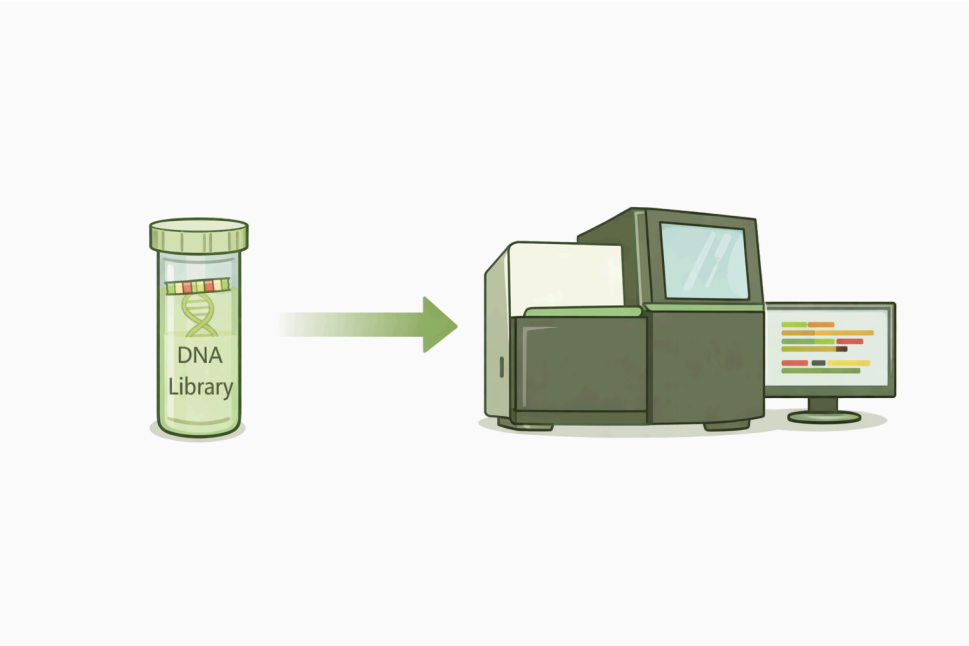
- Polyphenols and polysaccharides interfere with enzymes and make DNA sticky
- DNA quality varies between samples, tissues, methods and people

Step 2: Library preparation



- DNA is sheared, adapters are ligated, fragments are amplified by PCR
- Degraded DNA gives **uneven coverage**
- PCR amplifies some fragments more than others: **PCR bias**

Step 3: Sequencing



- Base calls get less accurate towards the end of the read
- Errors accumulate over cycles
- Lanes and runs differ in quality

Plant genomes are special

QC tools are **generic**. They don't know your data comes from a plant.

- Large genomes, high repeat content
- Polyploidy and high heterozygosity
- GC content that varies between species

So GC plots can look unusual and duplication can be high, even for good data.

QC tools show you *what* looks unusual. You need biological knowledge to decide *why*.

The raw reads

From the sequencer we get the raw data in FASTQ file format. Each read takes four lines:

```
@A00123:45:HKLMNDSX3:2:1101:10004:1000 1:N:0:ACGTACGT
GATCCTAGCAATGGTCGTACCGTTAGCTTGAACCTAGGATCAGTTCAAGC
+
FFFFFFFFF:FFFFFFFFFFF:FFF,FFFFFFF:FFFF:FF,F:F:,#:,
```

Line	Content
1	@ + read name: instrument, run, flow cell, lane, tile, position; 1 = read 1 of the pair
2	The sequence
3	+ (sometimes the name again)
4	One quality character per base

Quality scores

Phred score

$$Q = -10 \times \log_{10}(P_{error})$$

Q	Error rate	Accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1,000	99.9%
40	1 in 10,000	99.99%

One character per base

$$Q = \text{ASCII code} - 33$$

Character	ASCII	Q
#	35	2
,	44	11
:	58	25
F	70	37

FastQC

What FastQC does

- Reads a FASTQ file: millions of reads, each with a quality score per base
- Runs a series of standard tests, called modules
- Writes an HTML report with a figure or table per module

FastQC does not know your experiment or your plans. The interpretation and decisions are up to you.

Pass, warn, fail

Every module gets a label: ✅ pass, ⚠️ warn, ❌ fail.

- A fail does not mean the data is unusable
- A pass does not mean the data is perfect
- Thresholds are generic: they assume random, whole-genome DNA

Treat the flags as signals to investigate, not as decisions.

Basic statistics

Basic Statistics

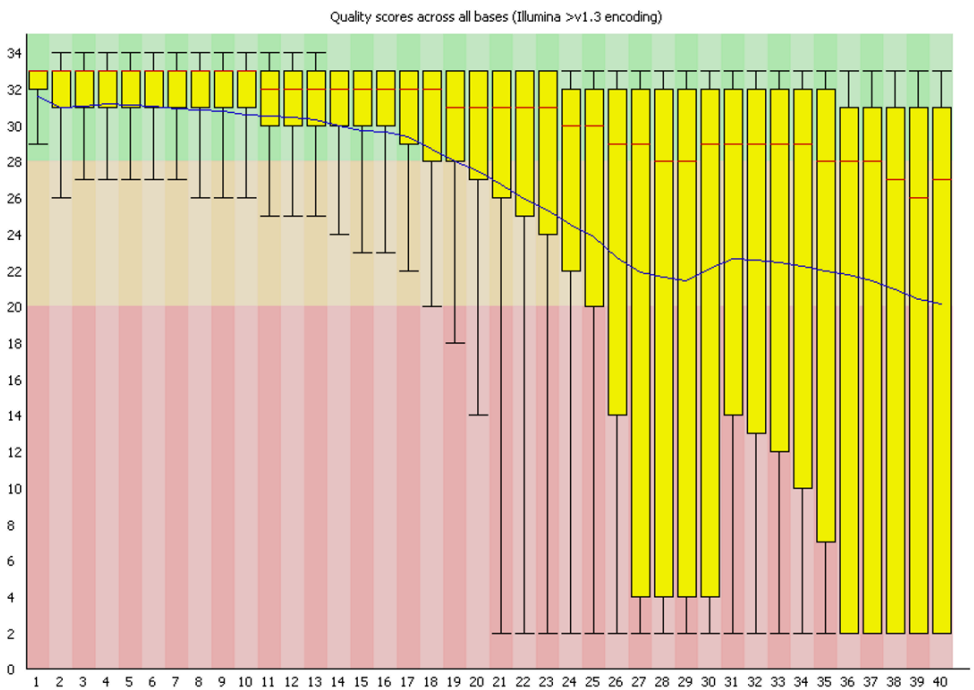
Measure	Value
Filename	good_sequence_short.txt
File type	Conventional base calls
Encoding	Illumina 1.5
Total Sequences	250000
Total Bases	10 Mbp
Sequences flagged as poor quality	0
Sequence length	40
%GC	45

Shows file name, number of reads, read length, GC %.

Check you are looking at the right sample, and compare read numbers between files.

Never raises a warning.

Per base sequence quality

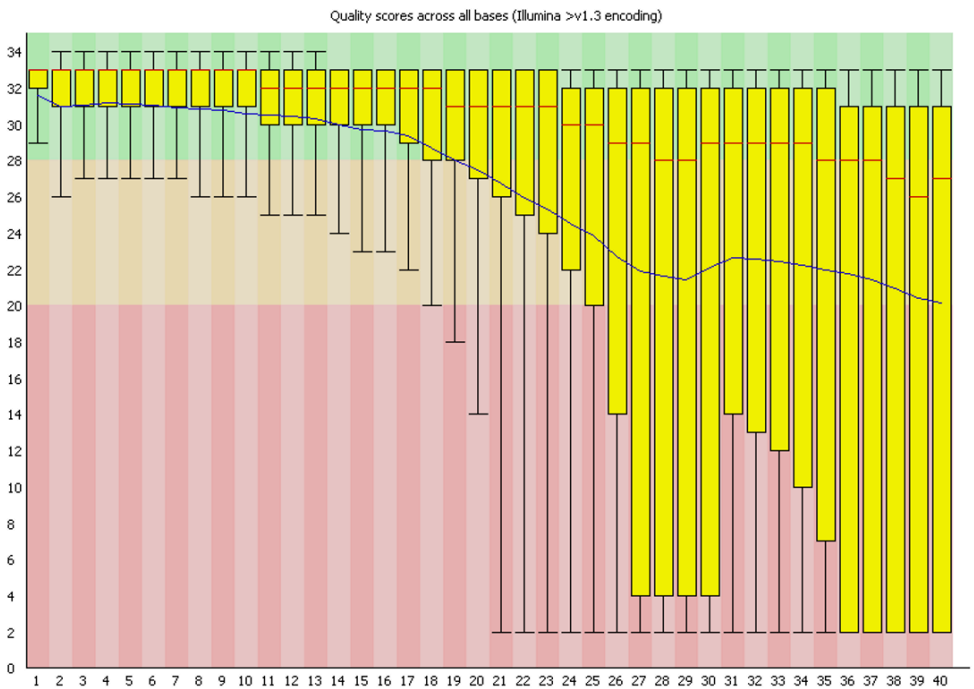


Shows quality scores at each position in the read.

We expect high quality at the start, a slow decline towards the end.

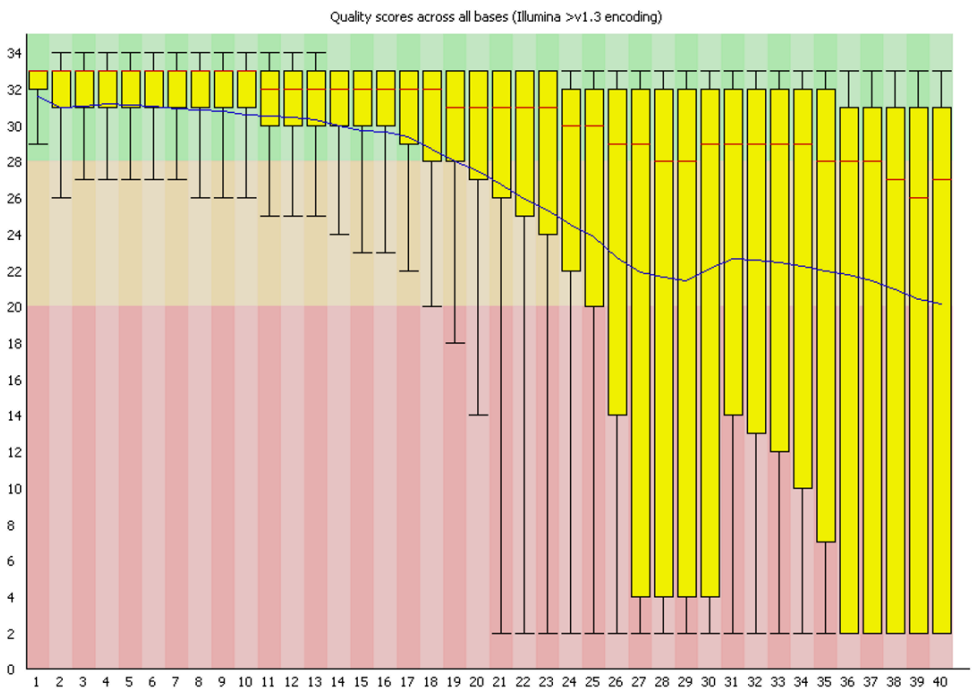
Warning signs include a sharp drop, large parts of the read in the red, big differences between samples.

Your turn: Pause the video



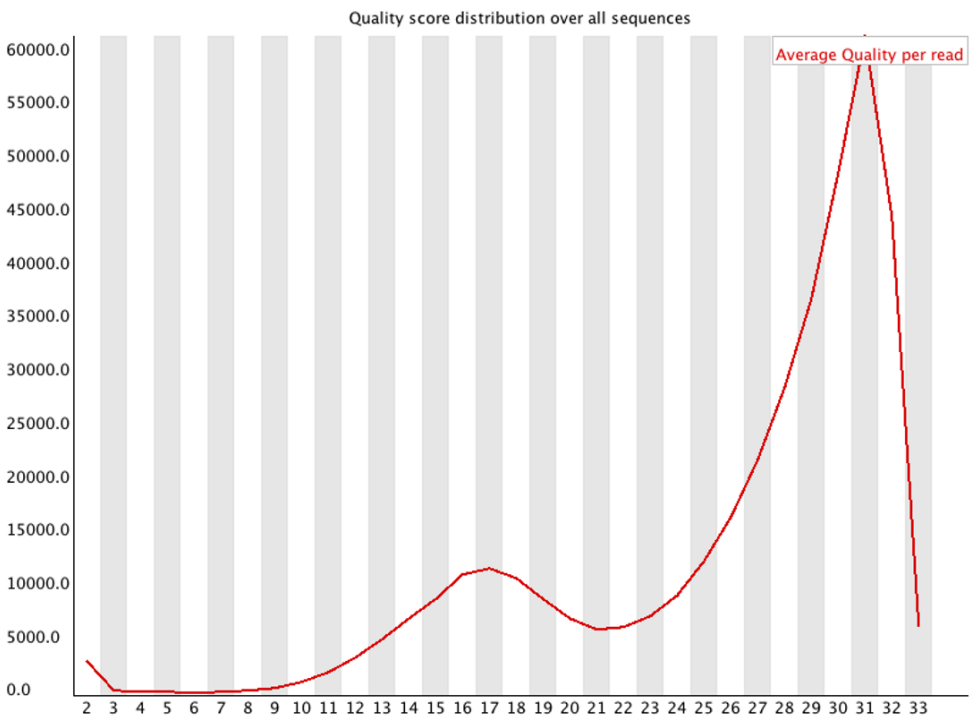
1. From which position does the quality drop?
2. Would you use these reads as they are?

The answer



1. The median (red line) stays above 26, but the mean (blue line) drops below 28 around position 19. From position 26 at least a quarter of the reads are in the red zone.
2. Not as they are. **Trim** the low-quality ends, then run FastQC again.

Per sequence quality scores

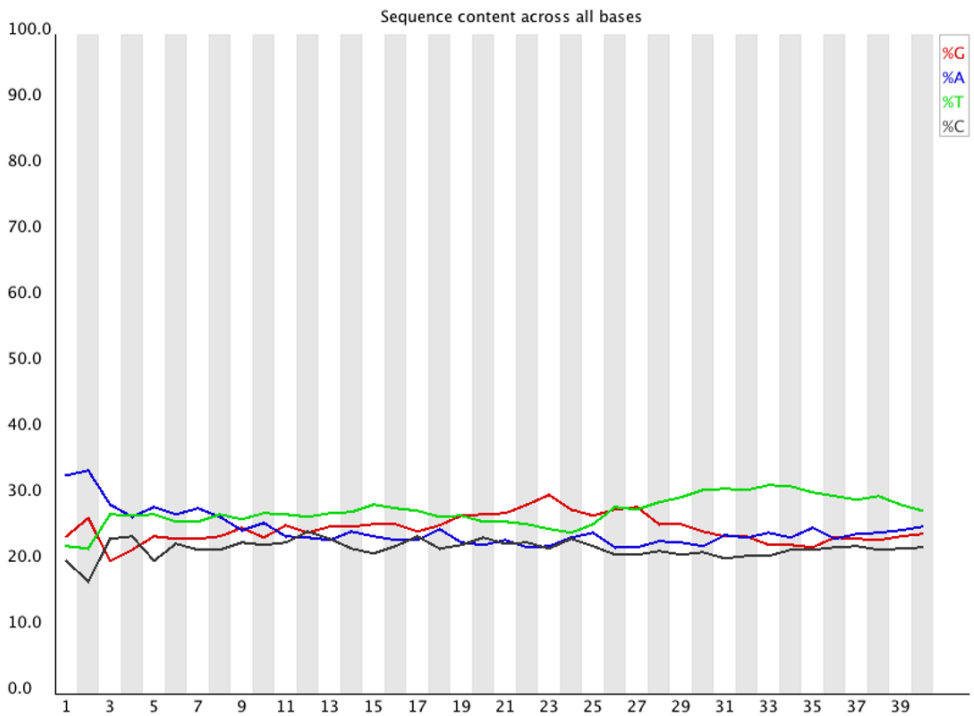


Shows the mean quality **per read**, not per position.

We expect one sharp peak at high quality.

Warning signals include a large share of low-quality reads, or a second peak. Points to library or sequencing problems, or poor starting material.

Per base sequence content

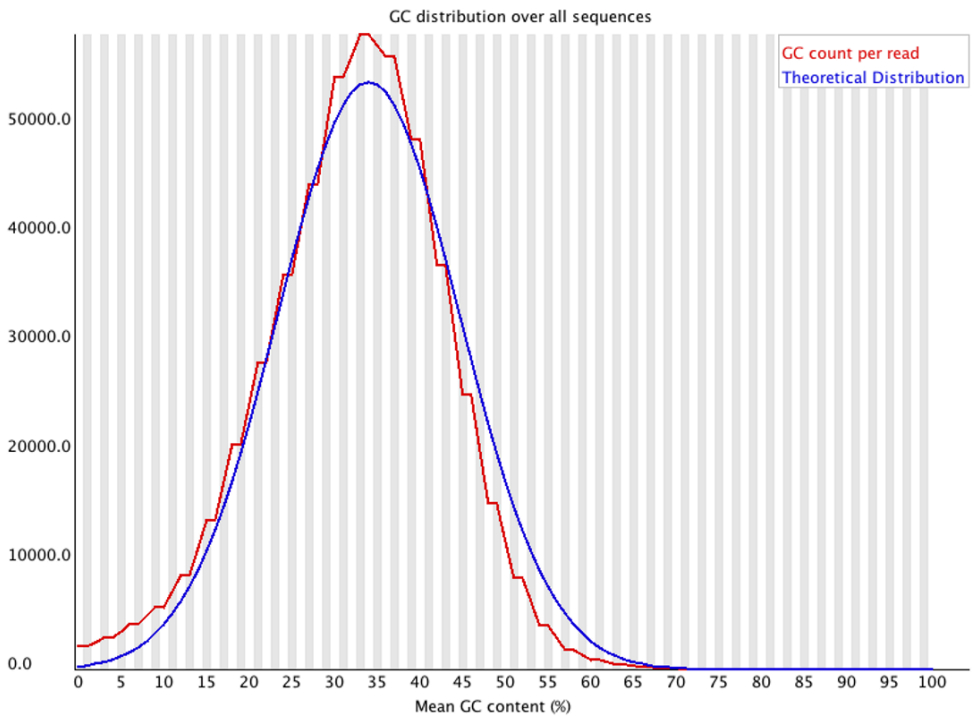


Shows the share of A, C, G and T at each position.

We expect flat lines for random genomic DNA.

But many library types are biased at the start of the read, RNA-seq always. Ask whether the pattern is expected for your experiment.

Per sequence GC content

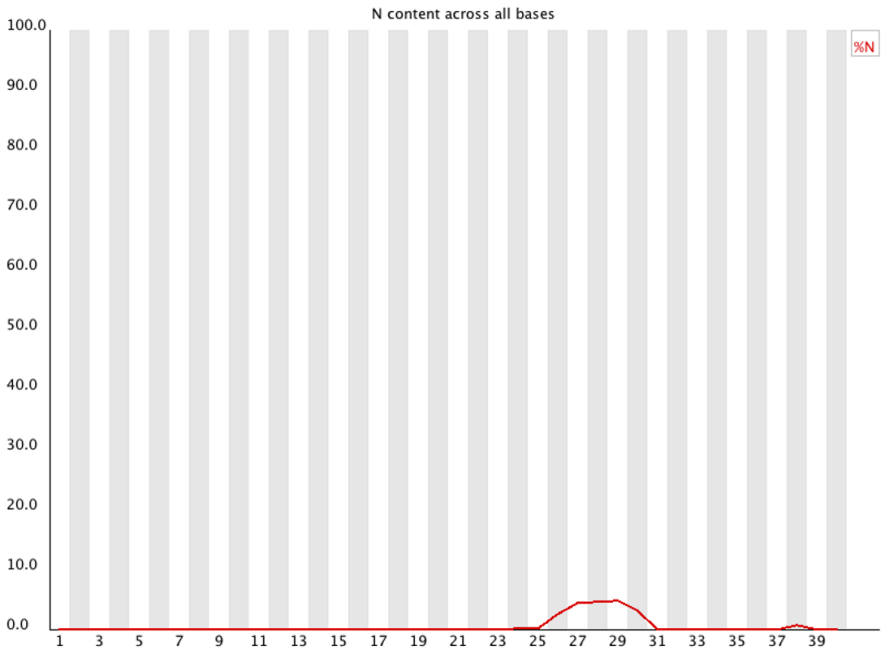


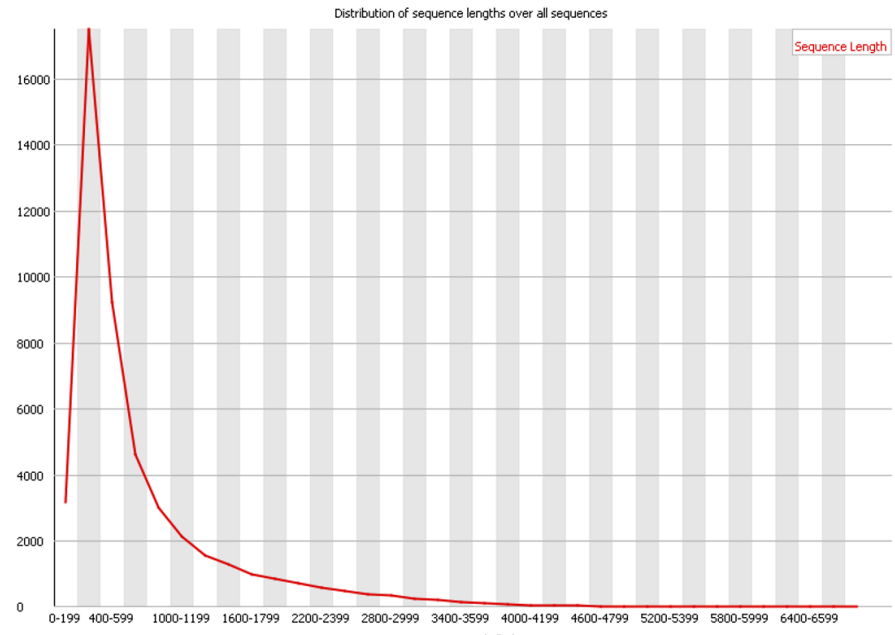
It shows GC content of the reads, against a theoretical curve.

We expect roughly a normal distribution, peaking at the genome's GC content.

Warning signals include a second peak or a shifted sample: contamination. In plants also repeats or genome composition. Always compare samples with each other.

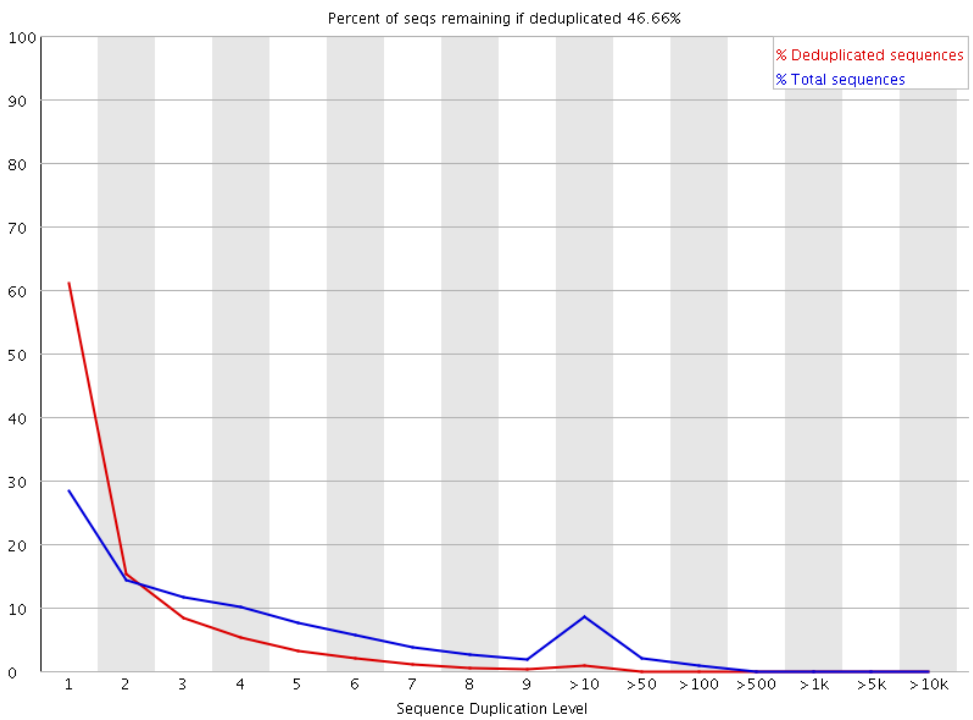
N content and read length





- N content: the sequencer could not call the base. A few at the end of reads is normal.
- Read lenght: Illumina reads have one length; PacBio and Nanopore vary. After trimming, lengths vary too.

Sequence duplication levels



It shows how often identical sequences occur.

We expect most sequences to occur once in a diverse library.

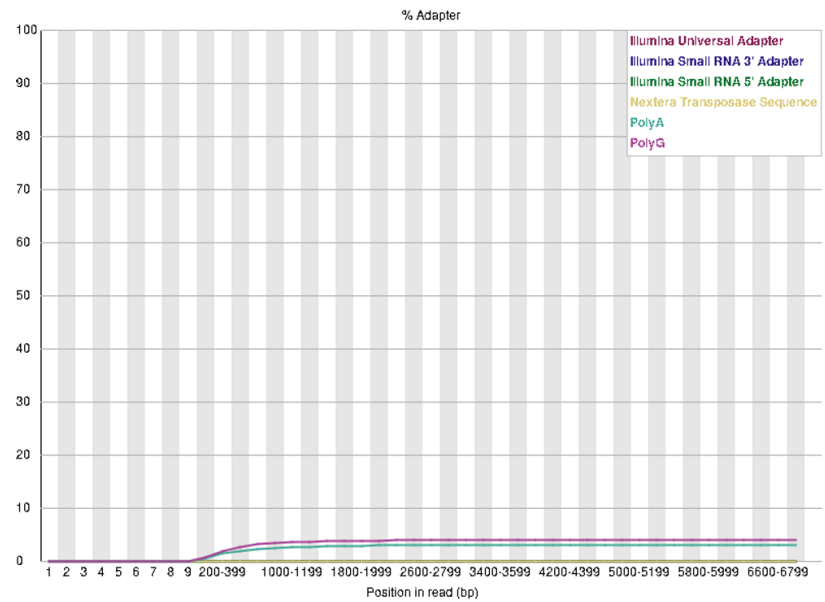
High duplication comes from PCR bias, low library complexity or very deep sequencing. In plants also conserved and repetitive regions. Normal in RNA-seq.

Overrepresented sequences and adapters

Overrepresented sequences

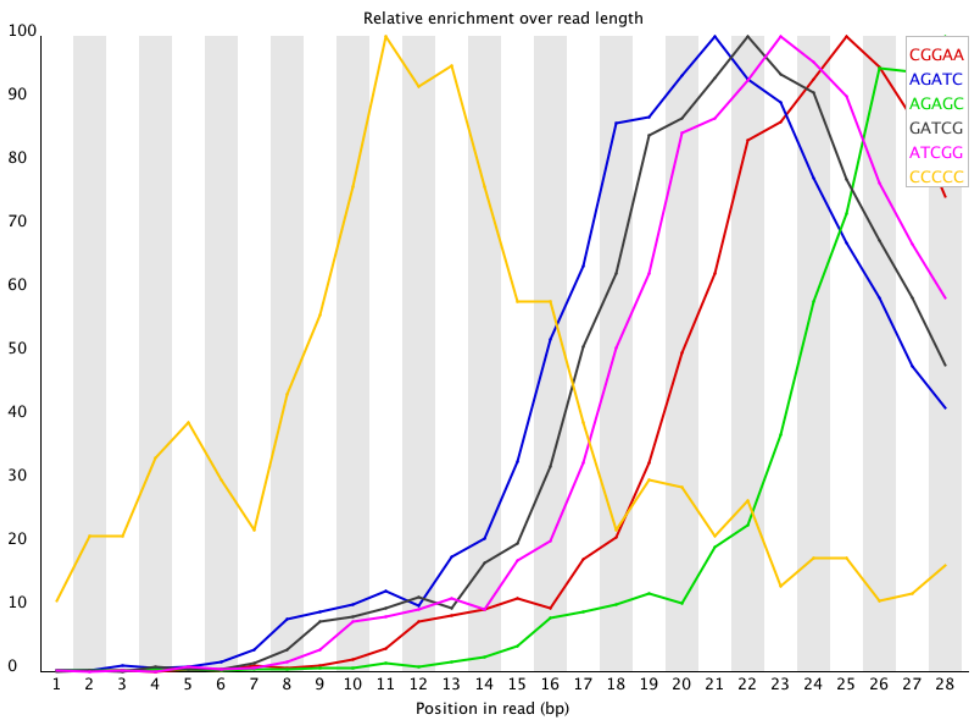
Sequence	Count	Percentage	Possible Source
AGAGTTTTATCGCTTCCATGACGCAGAAGTTAACACTTTC	2065	0.5224039181558763	No Hit
GATTGGCGTATCCAACCTGCAGAGTTTTATCGCTTCCATG	2047	0.5178502762542754	No Hit
ATTGGCGTATCCAACCTGCAGAGTTTTATCGCTTCCATGA	2014	0.5095019327680071	No Hit
CGATAAAAATGATTGGCGTATCCAACCTGCAGAGTTTTAT	1913	0.4839509420979134	No Hit
GTATCCAACCTGCAGAGTTTTATCGCTTCCATGACGCAGA	1879	0.47534961850600066	No Hit
AAAAATGATTGGCGTATCCAACCTGCAGAGTTTTATCGCT	1846	0.4670012750197325	No Hit
TGATTGGCGTATCCAACCTGCAGAGTTTTATCGCTTCCAT	1841	0.46573637449150995	No Hit
AACCTGCAGAGTTTTATCGCTTCCATGACGCAGAAGTTAA	1836	0.46447147396328753	No Hit
GATAAAAATGATTGGCGTATCCAACCTGCAGAGTTTTATC	1831	0.4632065734350651	No Hit
ATGATTGGCGTATCCAACCTGCAGAGTTTTATCGCTTCCA	1779	0.45005160794155147	No Hit
AAATGATTGGCGTATCCAACCTGCAGAGTTTTATCGCTTC	1779	0.45005160794155147	No Hit
AATGATTGGCGTATCCAACCTGCAGAGTTTTATCGCTTCC	1760	0.4452449859343061	No Hit
AAAATGATTGGCGTATCCAACCTGCAGAGTTTTATCGCTT	1729	0.4374026026593269	No Hit
CGTATCCAACCTGCAGAGTTTTATCGCTTCCATGACGCAG	1713	0.43335492096901496	No Hit
ATCCAACCTGCAGAGTTTTATCGCTTCCATGACGCAGAAG	1708	0.43209002044079253	No Hit
CAGAGTTTTATCGCTTCCATGACGCAGAAGTTAACACTTT	1684	0.42601849790532476	No Hit
TGCAGAGTTTTATCGCTTCCATGACGCAGAAGTTAACACT	1668	0.4219708162150128	No Hit

Adapter Content



- Overrepresented means single sequences are much more common than expected. Could be adapters, primers, repeats, or real biology.
- Adapter content shows which adapter is present, and how much. Tells you what to trim.

K-mer content (optional)



A k-mer is a short sequence of length k, for example the 7-mer `ATCGTGA`.

FastQC looks for k-mers enriched at certain positions. Enrichment at the start or end of reads usually means adapters, primers or library artefacts.

Only 2 % of the reads are checked.

After QC

Three possible outcomes

QC helps you to decide what to do next with your data.

1. Use as is: minor or expected warnings

2. Process: trim adapters or low-quality ends, then run FastQC again

3. Investigate: serious problems may mean re-preparing, resequencing or excluding the sample

QC in a reproducible workflow

- Run QC the same way on all samples
- Record which issues you saw
- Document what you did about them, and why

Many samples: MultiQC

Dozens of samples × 2 reads × FastQC before and after trimming: too many reports to read one by one.

MultiQC collects them into one report:

- Samples side by side
- Outliers stand out
- Systematic problems across samples become visible

You will use FastQC, MultiQC and fastp in the exercises.

Summary

- QC protects your biological conclusions from technical problems
- Remeber problems can start long before the FASTQ file: during extraction, library preparation, sequencing
- FastQC flags are signals, not verdicts; interpret them for *your* data and organism
- After QC: use, trim and re-check, or investigate
- Document what you checked and what you did

FastQC figures and module descriptions adapted from the FastQC documentation, Babraham Bioinformatics.